

Quiz Questions + Answers

Link to Quiz:

<https://docs.google.com/forms/d/e/1FAIpQLSd-io-02OgCpVMKRZcxDjG0ETpskGpmbfkzGIAUTwzxm7rTiA/viewform>

1). If cameras store light linearly, but our eyes perceive it logarithmically, how do we store an image that actually looks natural to a human?

Answer: Logarithmically, with higher sensitivity to changes in darker areas (shadows) than bright areas (highlights).

Reasoning: While cameras record linearly, human perception is logarithmic, leading to better differentiation in shadows. This mismatch necessitates gamma correction.

2). Why is Gamma Correction applied to raw linear camera data?

Answer: To compensate for a human's logarithmic perception of brightness.

Reasoning: To mimic natural perception, we apply the inverse (exponential/power law) to linear data so effects cancel out for the viewer.

3). What is the mathematical formula of Gamma Correction?

Answer: $s = C \cdot r^\gamma$

Reasoning: $s = C \cdot r^\gamma$ (r normalized 0–1), where γ controls the curve shape.

4). From the Gamma Correction in the previous equation, what does the 'r' stand for?

Answer: Input pixel value (normalised between 0 and 1).

Reasoning: 'r' is the input pixel value (normalised between 0 and 1).

5). A primary characteristic of HDR images compared to LDR (Typically 8-bit):

Answer: They encode a much wider luminance range, often using 10-12 bits per channel.

Reasoning: HDR has a wider span and more bits to avoid stepping/banding.

6). What does gamma decoding do?

Answer: Reverses gamma compression by raising values to a power, restoring linear light data.

Reasoning: Decoding reverses encoding to get back to linear space for computation/rendering.

7). The key conceptual difference between gamma correction and tone mapping is:

Answer: Gamma correction adjusts linear data to perceptual space, while tone mapping compresses HDR into displayable LDR.

Reasoning: Gamma fixes perception mismatch in SDR tone mapping solves dynamic range mismatch between HDR capture and LDR display.

8). How does light work?

Answer: Linearly, brightness scales proportionally, so 10,000 lumens is twice 5,000, and raw pixel values reflect this intensity.

Reasoning: Light is linear, measured in lumens (total emitted light) which feeds into luminance (intensity per unit area). Cameras capture this linearly, pixel values in raw images are directly proportional to the linear behaviour of light.

9). A typical tone mapping operator applies a non-linear curve that is:

Answer: Mostly linear in darker tones, then compresses highlights to avoid clipping.

Reasoning: TMOs preserve detail in shadows/midtones while rolling off highlights.

10). What is luminance?

Answer: A measure of light intensity per unit of projected area.

Reasoning: Lumens is the total emitted light feeding into luminance (intensity per unit projected area), and cameras capture pixel values linearly proportional to this.